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Studierichting: Informatica
Oefeningenreeks 6E nr. 11.2

$$f(x) = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32 = (x-2)^5 \text{ (dus kritiek punt bij } x = 2)$$

$$f'(x) = 5x^4 - 40x^3 + 120x^2 - 160x + 80$$

$$f''(x) = 20x^3 - 120x^2 + 240x - 160$$

$$f'''(x) = 60x^2 - 240x + 240$$

$$f^{IV}(x) = 120x - 240$$

$$f^V(x) = 120$$

$$f'(2) = 0$$

$$f''(2) = 0$$

$$f'''(2) = 0$$

$$f^{iv}(2) = 0$$

$$f^v(2) = 120 \neq 0$$

Dus $n = 4$, even $\Rightarrow$ zadelpunt in $(2, f(2))$

References